The ϕ^4 field as a test of EM–BP

Giovanni Mantovani

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1 The model: why a snapshot is a Markov chain

The one-dimensional stochastic Allen–Cahn equation (we write Θ for the temperature),

$$\partial_\tau \phi = \partial_x^2 \phi - V'(\phi) + \sqrt{2\Theta}\,\eta(x,\tau), \qquad V(\phi) = \tfrac{1}{4}(\phi^2-1)^2,$$

with η space–time white noise, is the gradient flow of $H[\phi] = \int \left[\tfrac{1}{2}(\partial_x\phi)^2 + V(\phi)\right] \mathrm{d}x$ shaken at temperature Θ. Its stationary law is therefore the Gibbs measure $p[\phi] \propto e^{-H[\phi]/\Theta}$: domains near ± 1 separated by kinks, long when cold (Figure 1c).

Markov in space, exactly. On a lattice of spacing h the energy couples only neighbours, $H_h = \sum_i \left[(\phi_{i+1}-\phi_i)^2/2h + hV(\phi_i)\right]$, so

$$p(\phi_1,\dots,\phi_L) \;\propto\; \prod_i \exp\Big(-\frac{(\phi_{i+1}-\phi_i)^2}{2\Theta h} - \frac{h\,V(\phi_i)}{\Theta}\Big),$$

a product of nearest-neighbour factors: a snapshot is a Markov chain in the site index, with no approximation. Its kernel comes from the transfer operator $\mathsf{T}(\phi,\phi') = e^{-hV(\phi)/2\Theta} e^{-(\phi'-\phi)^2/2\Theta h} e^{-hV(\phi')/2\Theta}$ and its Perron pair $\mathsf{T}\psi_0 = \lambda_0\psi_0$:

$$K(\phi' \mid \phi) = \frac{\mathsf{T}(\phi,\phi')\,\psi_0(\phi')}{\lambda_0\,\psi_0(\phi)} = \frac{\mathcal{N}(\phi';\phi,\Theta h)\;g(\phi')}{Z(\phi)}, \qquad g = \psi_0\, e^{-hV/2\Theta}, \qquad \pi = \psi_0^2,$$

a Gaussian step re-weighted at the arrival point (a "tilted walk"). So BP is the transfer-matrix computation and the exact score is available (grid BP, exact to 1.3×10^{-12}). The drift is non-linear, $\mathbb{E}[a'-a \mid a] = \gamma(\log Z)'(a)$ on the standardised field $a = \phi/s$ with step variance $\gamma = \Theta h/s^2$, so the paper's kernel $a' = \rho a + \varepsilon$ (linear drift) is misspecified. The correlation length $\xi = 1/\log(\lambda_0/\lambda_1)$ grows like $1/h$ under refinement and like $e^{E_{\mathrm{kink}}/\Theta}$ as the field cools; there is no transition in one dimension.

2 The estimator: your pair energy, learned through the noise

Model. A Gaussian bond plus a site energy, five parameters, with the truth in the family ($\gamma = \Theta h/s^2$, $f = (h/2\Theta)V(sa)$+const):

$$W_\theta(a,a') = \frac{(a'-a)^2}{2\gamma} + f_\theta(a) + f_\theta(a'), \qquad f_\theta(a) = \sum_{m=1}^{4} c_m \Big(\frac{a}{2}\Big)^m.$$

We call this arm the *site energy*. Its kernel is the ground-state transform of $e^{-f_\theta(a)}\mathcal{N}(a';a,\gamma)\,e^{-f_\theta(a')}$, and a fit reads as physics: $\hat\Theta = \hat\gamma s^2/h$, $\hat V(\phi) = (2\hat\Theta/h)\hat f(\phi/s)$+const.

Learning. From noisy data only, $\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}$. Fisher's identity makes EM your Boltzmann learning with a posterior in place of the data:

$$\nabla_\theta \log p_\theta(\mathbf{x}) = -\,\mathbb{E}_{p_\theta(\mathbf{a}|\mathbf{x})}\Big[\textstyle\sum_i \nabla_\theta W_\theta(a_i,a_{i+1})\Big] + \mathbb{E}_{p_\theta(\mathbf{a})}\Big[\textstyle\sum_i \nabla_\theta W_\theta(a_i,a_{i+1})\Big].$$

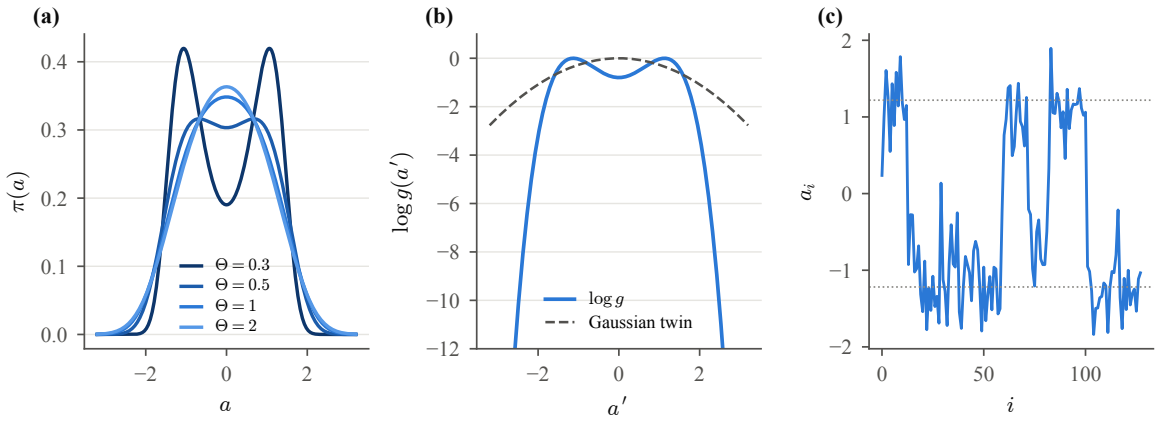


Figure 1: (a) Stationary density of the standardised field at the four temperatures of the sweep: two humps when cold. (b) The log-tilt $\log g$ against the Gaussian tilt of the covariance twin: bimodal, which no linear-Gaussian kernel can express. (c) An exact snapshot of 128 sites at $\Theta = 0.3$, $h = 1$; dotted lines mark the wells.

Table 1: Risk divided by the linear floor (the Gaussian closure scores 1) at $M = 2048$, mean over sixteen seeds. The first four columns are the temperature sweep at matched correlation length, the last two refine the base cell at fixed physical length. A dash: all sixteen fits failed. Bold: smallest in its column. The full table, with $M = 128$, is in the backup, Section B3.

M	estimator	$\Theta = 0.3$	0.5	1	2	$h = 0.5$	$h = 0.25$
2048	site energy (5)	**0.033**	**0.072**	**0.141**	**0.169**	**0.040**	**0.032**
	spline tilt (25)	0.060	0.134	0.280	0.429	0.073	0.068
	Legendre tilt (7)	0.074	0.092	0.165	0.243	0.111	–
	MDN	0.228	0.422	0.737	0.837	0.310	0.420
	mixture (paper)	1.004	1.004	1.014	1.021	1.010	1.016
	window, ISM	0.552	1.284	2.457	3.559	0.776	1.196
	transformer, ISM	1.354	2.706	2.440	3.237	1.368	1.020
	window, DSM	0.484	1.174	2.287	3.334	0.701	1.030
	transformer, DSM	0.533	1.225	2.156	3.043	0.622	0.780

The E-step is exact BP on each noisy sequence; the M-step refits θ to the expected pair counts. The score then follows by Tweedie, $\nabla_{\mathbf{x}} \log P_t(\mathbf{x}) = -(\mathbf{x} - e^{-t}\mathbb{E}[\mathbf{a} \mid \mathbf{x}])/\Delta_t$, with $\mathbb{E}[\mathbf{a} \mid \mathbf{x}]$ from BP with the learned kernel.

3 The experiments

Six cells. A temperature sweep $\Theta \in \{0.3, 0.5, 1, 2\}$ with h solved to hold $\xi \approx 6$ sites, and a refinement $h \in \{1, \frac{1}{2}, \frac{1}{4}\}$ of the base cell $\Theta = 0.3$ at fixed physical length ($L = 32, 64, 128$). **Arms.** EM–BP with five kernels: site energy (5 numbers), spline tilt (25), Legendre tilt (7), the paper's mixture (25), an MDN (1,516); a window CNN and a transformer, trained on one noisy copy per sequence (ISM) or on clean targets (DSM, more information); the Gaussian closure, i.e. the best linear estimator ("linear floor"); the exact oracle. **Protocol.** $M \in \{32, 128, 512, 2048\}$ sequences, sixteen seeds, predictions registered before each run. **Metric.** $R = \left(\sum \|\hat{S} - S\|^2 / \sum \|S\|^2\right)^{1/2}$ over twelve noise levels $t \in [0.05, 3]$, reported as R divided by the linear floor: 1 is the Gaussian closure.

4 Results

1. **The five-parameter site energy is best on every cell** (Table 1). At $M = 2048$ its error is 0.032–0.169 of the Gaussian closure's, 0.41–0.57 of the 25-number spline's, and the best network

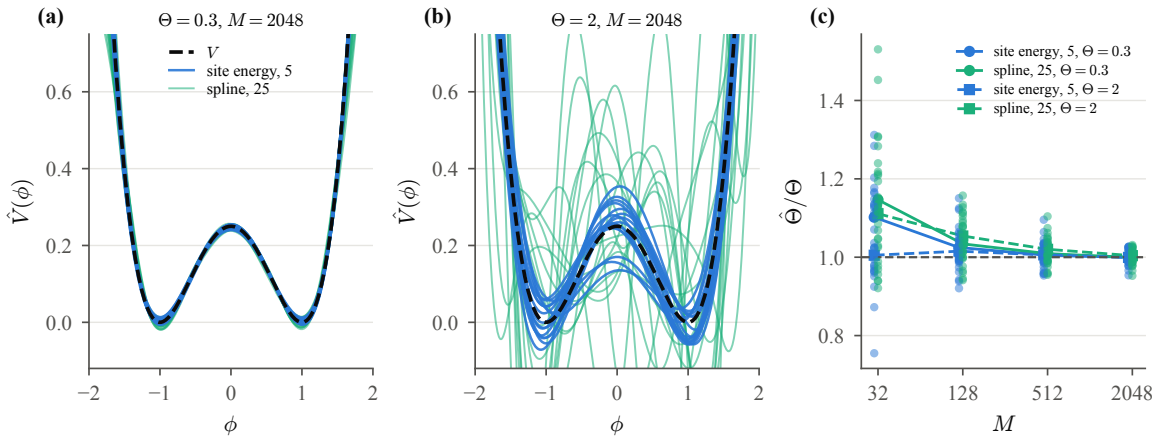


Figure 2: The potential read off the kernels learned through noise, one curve per seed (sixteen), against the true double well (dashed), $M = 2048$. (a) Base cell. (b) $\Theta = 2$, where the barrier is small against the thermal energy. (c) Fitted over true temperature against M, every seed as a point.

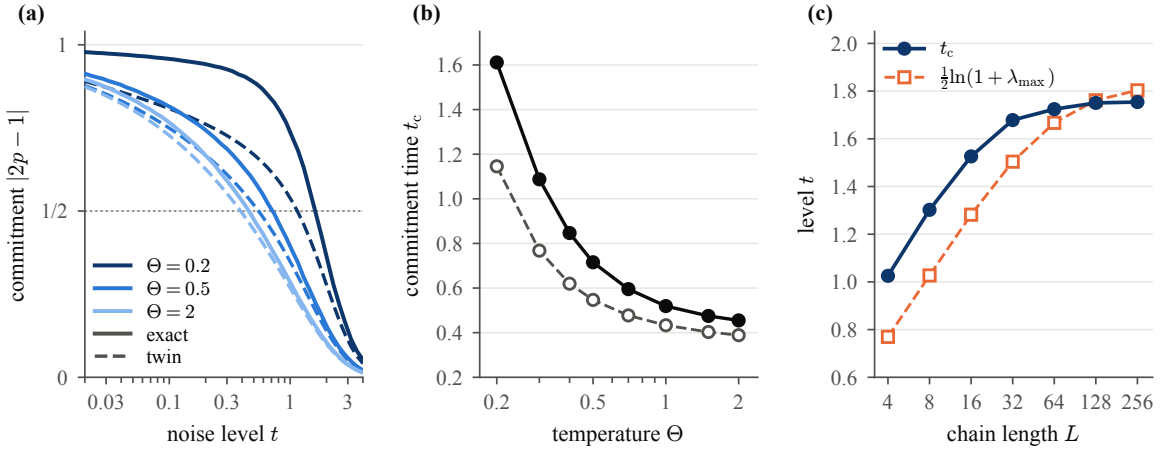


Figure 3: Speciation. (a) Site-averaged commitment $|2p - 1|$, $p = P(a_i > 0 \mid \mathbf{x})$, against t: exact posterior (solid) and its Gaussian twin with the same covariance (dashed). (b) The level t_c where it crosses $\frac{1}{2}$, against Θ. (c) At $\Theta = 0.2$, t_c against the chain length, beside $\frac{1}{2}\ln(1 + \lambda_{\max})$.

is 14–24 times worse.

2. **The kernel family is the modelling choice.** The paper's mixture learns nothing beyond the Gaussian closure (1.004–1.021 times it): a linear drift cannot hold a domain.
3. **The fit is the physics** (Figure 2): $\hat{\Theta}$ within 10% in 96 of 96 fits, barrier within 20% in 78 of 80; samples drawn from it match the exact kink density within 2.1%.
4. **Your refinement prediction holds.** From $h = 1$ to $\frac{1}{4}$ the window CNN goes from 0.55 to 1.20 times the floor (radius re-chosen per lattice); the site energy stays at 3–4%.
5. **Speciation** (Figure 3): colder fields commit earlier (t_c from 1.61 at $\Theta = 0.2$ to 0.46 at $\Theta = 2$), and earlier than their Gaussian twin (by 0.47 to 0.07 in t).
6. **Caveats** (backup, B7). BP is slower per sequence than a network (12.5× the transformer, eight threads, 401 points; 101 points lose nothing here and make BP 6.2× faster than on 401). EM and the DSM networks stop on the exact score of a validation set (a likelihood rule for EM gives 4–23%); the ISM networks do not. A neural kernel wins only off-class (backup, B6).

The separate backup has your email point by point, the derivations, the full table, cost and refinement, the stress tests, what we do not claim, and the answers to the 25 September questions.